

MODULAR ARITHMETIC

Congruence $a \equiv b \pmod{n} \iff n \mid (a - b)$. Division by b : multiply by b^{-1} ; valid iff $\gcd(b, n) = 1$.
Euclid/Bézout $a = qb + r$; $\gcd(a, b) = \gcd(b, a \bmod b)$; $ax + by = \gcd(a, b)$. In particular,

$$\gcd(a, N) = 1 \iff \exists x, y \in \mathbb{Z} : ax + Ny = 1, \quad a^{-1} \equiv x \pmod{N}.$$

Extended Euclid table Start $(r_0, s_0, t_0) = (a, 1, 0)$, $(r_1, s_1, t_1) = (b, 0, 1)$; for $q_i = \lfloor r_{i-1}/r_i \rfloor$,

$$(r_{i+1}, s_{i+1}, t_{i+1}) = (r_{i-1}, s_{i-1}, t_{i-1}) - q_i(r_i, s_i, t_i).$$

Last nonzero row $= (\gcd, x, y)$.

Square-and-multiply For bits of e from MSB to LSB: initialize $r = 1$; each bit do $r \leftarrow r^2 \bmod n$; if bit 1, $r \leftarrow ra \bmod n$. Computes $a^e \bmod n$ in $O(\log e)$.

Euler totient If $n = \prod p_i^{e_i}$,

$$\varphi(n) = n \prod_{p|n} (1 - 1/p) = \prod p_i^{e_i-1} (p_i - 1).$$

$\varphi(p) = p - 1$; $\varphi(p^k) = p^{k-1}(p - 1)$; $\varphi(pq) = (p - 1)(q - 1)$ for distinct primes; $\gcd(a, b) = 1 \Rightarrow \varphi(ab) = \varphi(a)\varphi(b)$.

Fermat/Euler

$$p \text{ prime, } p \nmid a \Rightarrow a^{p-1} \equiv 1 \pmod{p},$$

$$\gcd(a, n) = 1 \Rightarrow a^{\varphi(n)} \equiv 1 \pmod{n}.$$

Reduce exponents mod $p - 1/\varphi(n)$ only for coprime base. $a^{n-1} \not\equiv 1 \pmod{n} \Rightarrow n$ composite (not conversely).

CRT Pairwise coprime n_i ; $N = \prod n_i$, $N_i = N/n_i$, $u_i = N_i^{-1} \bmod n_i$:

$$x \equiv \sum_i a_i N_i u_i \pmod{N} \iff x \equiv a_i \pmod{n_i}.$$

GROUPS AND ORDER

$\mathbb{Z}_n^*$ has order n ; $\mathbb{Z}_n^* = \{a : \gcd(a, n) = 1\}$ has order $\varphi(n)$. Identity is 0 (additive), 1 (multiplicative).

$$\text{ord}(g) = \min\{r > 0 : g^r = e\}, \quad \text{ord}(g) \mid |G|.$$

$$\text{ord}_{\mathbb{Z}_h}(a) = n / \gcd(a, n),$$

$$\text{ord}(g^k) = \text{ord}(g) / \gcd(\text{ord}(g), k).$$

Cyclic $|G| = N$: $\varphi(N)$ generators; $\varphi(d)$ elements of order $d \mid N$. If $|G| = q$ prime, every $g \neq e$ has order q .

Find $\text{ord}(g)$ Factor N (usually $p - 1/\varphi(n)$); $r \leftarrow N$. For each prime $\ell \mid r$, while $g^{r/\ell} = e$, set $r \leftarrow r/\ell$. Return r .

Generator test If prime divisors of $N = |G|$ are ℓ_i ,

$$g \text{ generates } G \iff g^{N/\ell_i} \neq e \ \forall i.$$

PRIMES / MILLER-RABIN

$\pi(x) \sim x/\ln x$; density near x : $1/\ln x$ (or $2/\ln x$ among odd candidates).

One round For odd $n > 2$, write $\lfloor \frac{n-1}{2} \rfloor = 2^s d$ with d odd. Choose $2 \leq a \leq n-2$.

$\gcd(a, n) > 1 \Rightarrow$ COMPOSITE. Otherwise $x \leftarrow a^d \bmod n$. If $x \in \{1, n-1\} \Rightarrow$ PASS. Repeat $s-1$ times: $x \leftarrow x^2 \bmod n$; if $x = n-1 \Rightarrow$ PASS. If never reached, COMPOSITE.

Sequence: $a^d, a^{2d}, \dots, a^{2^{s-1}d}$. COMPOSITE is certain; PASS = probable prime. Fixed odd composite n :

$$\Pr[\text{all } t \text{ rounds pass}] \leq 4^{-t} = 2^{-2t}.$$

k -bit prime: sample $\lfloor 2^{k-1}, 2^k \rfloor$; force MSB/LSB 1; small-prime trial division; independent MR rounds.

SECURITY / SYMMETRIC CRYPTO

Correctness $\Pr[\text{Dec}_k(\text{Enc}_k(m)) = m] = 1$.

Perfect secrecy $\Pr[M = m \mid C = c] = \Pr[M = m]$; equivalently ciphertext distribution is message-independent.

Advantage $\text{Adv}_{\mathcal{A}} = |\Pr[b' = b] - 1/2|$. μ negligible iff $\forall c > 0$, eventually $\mu(\lambda) < \lambda^{-c}$. $2^{-\lambda}$: yes; λ^{-100} : no.

PRG $G : \{0, 1\}^\lambda \rightarrow \{0, 1\}^{\ell(\lambda)}$, $\ell > \lambda$; $|\Pr[D(G(s)) = 1] - \Pr[D(U_\ell) = 1]| \leq \text{negl}(\lambda)$. Hybrid bound: $|\Pr[H_0] - \Pr[H_t]| \leq \sum_i |\Pr[H_i] - \Pr[H_{i+1}]|$.

PRF/PRP PRF resembles random function (collisions possible); PRP is keyed bijection $\{0, 1\}^n \rightarrow \{0, 1\}^n$ (block cipher).

EAV/CPA EAV: challenge only. CPA: adaptive $\text{Enc}_k(\cdot)$ queries. Deterministic $\Rightarrow$ not CPA. CPA $\nRightarrow$ integrity.

OTP/stream $c = m \oplus z$, $m = c \oplus z$. OTP: $z = k$ uniform/secret, $|k| \geq |m|$, one use.

Reuse: $c \oplus c' = m \oplus m'$.

Feistel

$$\begin{aligned} L_i &= R_{i-1}, & R_i &= L_{i-1} \oplus F_i(R_{i-1}); \\ R_{i-1} &= L_i, & L_{i-1} &= R_i \oplus F_i(L_i). \end{aligned}$$

F_i need not invert; reverse subkeys.

AES Block 128. Key/rounds: 128/10, 192/12, 256/14. Round: SB $\rightarrow$ SR $\rightarrow$ MC $\rightarrow$ ARK; initial ARK; final omits MC. In $GF(2^m)$: add=XOR; multiply polynomials mod irreducible f . AES: $x^8 + x^4 + x^3 + x + 1$; GCM: $x^{128} + x^7 + x^2 + x + 1$.

DES/3DES DES: block 64, key 56, 16-round Feistel. 3DES-EDE: $c = E_{k_3}(D_{k_2}(E_{k_1}(m)))$; about 112-bit classical security.

ECB $c_i = E_k(m_i)$: deterministic, leaks equal blocks.

CBC fresh unpredictable IV; padding required:

$$c_0 = IV, \quad c_i = E_k(m_i \oplus c_{i-1}), \quad m_i = D_k(c_i) \oplus c_{i-1}.$$

Flip in c_i : garbles m_i , predictably flips same bit in m_{i+1} . No authentication.

CTR unique nonce per key; no padding:

$$z_i = E_k(N \parallel \langle i \rangle), \quad c_i = m_i \oplus z_i.$$

Nonce reuse repeats keystream; ciphertext bit flip = plaintext bit flip.

GCM = CTR + GHASH (AEAD)

$$H = E_k(0^{128}), \quad T = \text{MSB}_t(E_k(J_0) \oplus \text{GHASH}_H(A, C)).$$

Unique nonce; reuse breaks C+I. Verify tag before releasing plaintext.

RSA

Keygen distinct primes p, q :

$$n = pq, \quad \varphi(n) = (p-1)(q-1),$$

$$\gcd(e, \varphi(n)) = 1, \quad d = e^{-1} \bmod \varphi(n).$$

$\text{pk} = (n, e)$; $\text{sk} = (n, d)$ (store p, q for CRT).

$$\text{Enc}_{\text{pk}}(m) = m^e \bmod n, \quad \text{Dec}_{\text{sk}}(c) = c^d \bmod n.$$

If $\gcd(m, n) = 1$: $ed = 1 + t\varphi(n) \Rightarrow m^{ed} = m(m^{\varphi(n)})^t \equiv m$. Otherwise prove mod p, q , then CRT; still correct, but $\gcd(m, n) > 1$ reveals a factor.

CRT decryption $d_p = d \bmod (p-1)$, $d_q = d \bmod (q-1)$; $m_p = c^{d_p} \bmod p$, $m_q = c^{d_q} \bmod q$; recombine by CRT.

Factor from $\varphi(n)$ Let $S = n - \varphi(n) + 1 = p + q$:

$$p, q = (S \pm \sqrt{S^2 - 4n})/2.$$

Textbook RSA fails deterministic; $\text{Enc}(m_1)\text{Enc}(m_2) \equiv \text{Enc}(m_1 m_2) \pmod{n}$. Use OAEP/PSS. OAEP: $X = M \oplus G(r)$, $Y = r \oplus H(X)$; undo $r = Y \oplus H(X)$, $M = X \oplus G(r)$.

DH / ELGAMAL / HARD PROBLEMS

In $G = \langle g \rangle$, $|G| = q$: **DH** $A = g^a$, $B = g^b$, $K = B^a = A^b = g^{ab}$; use KDF. No authentication $\Rightarrow$ MITM.

DLP/CDH/DDH DLP: $g, g^x \mapsto x$; CDH: $g^a, g^b \mapsto g^{ab}$; DDH: distinguish (g^a, g^b, g^{ab}) from (g^a, g^b, g^c) . DLP $\Rightarrow$ CDH $\Rightarrow$ DDH (not converse generally).

ElGamal encryption Secret x , public $h = g^x$. Fresh y :

$$(c_1, c_2) = (g^y, mh^y), \quad m = c_2(c_1^{-1})^{-1}.$$

Reuse y : $c_2/c_2' = m/m'$.

Safe prime $p = 2q + 1$ primes; quadratic residues form order- q subgroup. $c = b^2 \not\equiv 1 \pmod{p} \Rightarrow \text{ord}(c) = q$.

Attack costs DLP brute force $O(q)$; BSGS $O(\sqrt{q})$ time/memory; rho-DLP $O(\sqrt{q})$ time/ $O(1)$ memory; rho-factor $O(\sqrt{p})$, p smallest factor; Index Calculus subexp. in $\mathbb{Z}_p^*$, not ECC.

Pollard-rho factoring $f(x) = x^2 + c \bmod n$; $x \leftarrow f(x)$, $y \leftarrow f(f(y))$; $d = \gcd(|x - y|, n)$. $d = 1$: continue; $1 < d < n$: factor; $d = n$: restart.

Pollard-rho DLP Collision $g^{a_i} h^{b_i} = g^{a_j} h^{b_j}$, $h = g^x$:

$$x = (a_j - a_i)(b_i - b_j)^{-1} \bmod q.$$

ELLIPTIC CURVES

$E : y^2 = x^3 + ax + b \pmod{p}$; require $4a^3 + 27b^2 \not\equiv 0 \pmod{p}$. For $P = (x_1, y_1)$, $Q = (x_2, y_2)$:

$$\lambda = \begin{cases} (y_2 - y_1)(x_2 - x_1)^{-1}, & P \neq \pm Q, \\ (3x_1^2 + a)(2y_1)^{-1}, & P = Q, \end{cases} \pmod{p}$$

$$x_3 = \lambda^2 - x_1 - x_2, \quad y_3 = \lambda(x_1 - x_3) - y_1 \pmod{p}.$$

$P + \mathcal{O} = P$; $-P = (x, -y)$; $P + (-P) = \mathcal{O}$. EC keys: $\text{sk} = x$, $\text{pk} = Q = xP$.

$P, Q = xP \mapsto x$; generic cost $O(\sqrt{q})$ for $\text{ord}(P) = q$.

HASHES / BIRTHDAY / MAC

$H : \{0, 1\}^* \rightarrow \{0, 1\}^n$. Generic work: preimage 2^n ; second preimage 2^n ; collision $2^{n/2}$.

Birthday For q uniform draws from N values:

$$\begin{aligned} P_{\text{coll}} &= 1 - \prod_{i=0}^{q-1} (1 - i/N) \\ &\approx 1 - e^{-q(q-1)/(2N)}, \end{aligned}$$

$$P_{\text{coll}} = 1/2 \Rightarrow q \approx 1.177\sqrt{N}.$$

n -bit hash $\Rightarrow \leq n/2$ collision-security bits.

HMAC

$$\text{HMAC}_K(m) = H((K' \oplus \text{opad}) \| H((K' \oplus \text{ipad}) \| m)).$$

$H(k \| m)$ may length-extend. Basic CBC-MAC: fixed length only; variable $\Rightarrow$ CMAC. Hash alone $\neq$ authentication.

DIGITAL SIGNATURES

RSA textbook (unsafe) $\sigma = m^d \bmod n$; verify $\sigma^e \equiv m \pmod{n}$. Forge: choose σ , set $m = \sigma^e \bmod n$. Use PSS.

ElGamal signature p prime; g generator; secret x , public $y = g^x \bmod p$. Choose secret fresh k with $\gcd(k, p-1) = 1$:

$$r = g^k \bmod p,$$

$$s = k^{-1}(H(m) - xr) \bmod (p-1),$$

$$g^{H(m)} \stackrel{?}{=} y^r r^s \pmod{p}.$$

r : mod p ; s : mod $p-1$.

DSA $q \mid (p-1)$ prime; $\text{ord}(g) = q \bmod p$; secret $x \in \mathbb{Z}_q^*$, public $y = g^x \bmod p$. Fresh secret $k \in \mathbb{Z}_q^*$:

$$r = (g^k \bmod p) \bmod q, \quad s = k^{-1}(H(m) + xr) \bmod q.$$

Retry if $r = 0$ or $s = 0$. Verify

$$w = s^{-1} \bmod q, \quad u_1 = H(m)w \bmod q, \quad u_2 = rw \bmod q,$$

$$v = (g^{u_1} y^{u_2} \bmod p) \bmod q \stackrel{?}{=} r.$$

Exponentiation: mod p ; r, s, w, u_1, u_2, v : mod q .

Repeated DSA nonce recovers key Let $h_i = H(m_i)$. Then

$$k = (h_1 - h_2)(s_1 - s_2)^{-1} \bmod q,$$

$$x = (s_1 k - h_1)r^{-1} \bmod q.$$

ECDSA: $g^k \mapsto kG$. ElGamal/DSA/ECDSA nonce: secret, unpredictable, never reused.

10-SECOND ERROR CHECK

Inverse exists? $\gcd = 1$. Order divides group size. Fermat/Euler base coprime. MR uses $n-1 = 2^d d$ with d odd. RSA uses $d = e^{-1} \bmod \varphi(n)$. CBC: unpredictable IV; CTR/GCM: unique nonce. Validate group/subgroup membership. ElGamal signature: mod $p-1$; DSA: mod q . Hash $\neq$ authentication; unauthenticated encryption is malleable.